



**FACULTY OF INFORMATION SCIENCE
CAMPUS PUNCAK PERDANA**

BACHELOR OF INFORMATION SCIENCE (HONS.)
SYSTEM MANAGEMENT (SI262)

ADVANCED WEB DESIGN DEVELOPMENT AND CONTENT MANAGEMENT
(IMS566)

GROUPING ASSIGNMENT :

LOST & FOUND SYSTEM
(UiTEMU)

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17 JULY 2026

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SHAH ALAM SELANGOR

ACKNOWLEDGEMENT

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1.0 INTRODUCTION

UiTEMU are Lost & Found System is one of the applications created specifically for UiTM Puncak Perdana students. This application was created to help make it easier for students to report the loss of their belongings. Students can also find their belongings again by just checking from this system.

The main purpose of this Lost & Found system is to make it easier for students to report the loss of their belongings without having to go to the office or search everywhere. Students can also report items found in the system and can be seen by everyone. Students who want to make a report need to log in to the system using their own email, and each item will be updated from time to time. Students can also see the dashboard about their activities in the system. Students can also see a list of 10 users who have made a lot of reports on finding items. There is also a bar graph of their activities.

In this UiTEMU as an admin, they can see the number of users lost items, found items and total claims. Admins also need to update what items have been claimed and resolved. Only admins can see recent lost items and recent found items. There is also a certificate for users who find items as a sign of appreciation to them. Admin can also download the monthly report of Lost & Found System activities to be used as a reference.

This system will ease and speed up the discovery and reporting process for students. With the facilities available, students no longer need to make a report to the office or police to seek help from them. All students can access including lecturers or everyone who is at UiTM. There are no restrictions or limits to them for each item.

2.0 GITHUB REPOSITORY LINK

<https://github.com/irdinanurin/UiTEMU>

3.0 ENTITY RELATIONSHIP DIAGRAM (ERD)

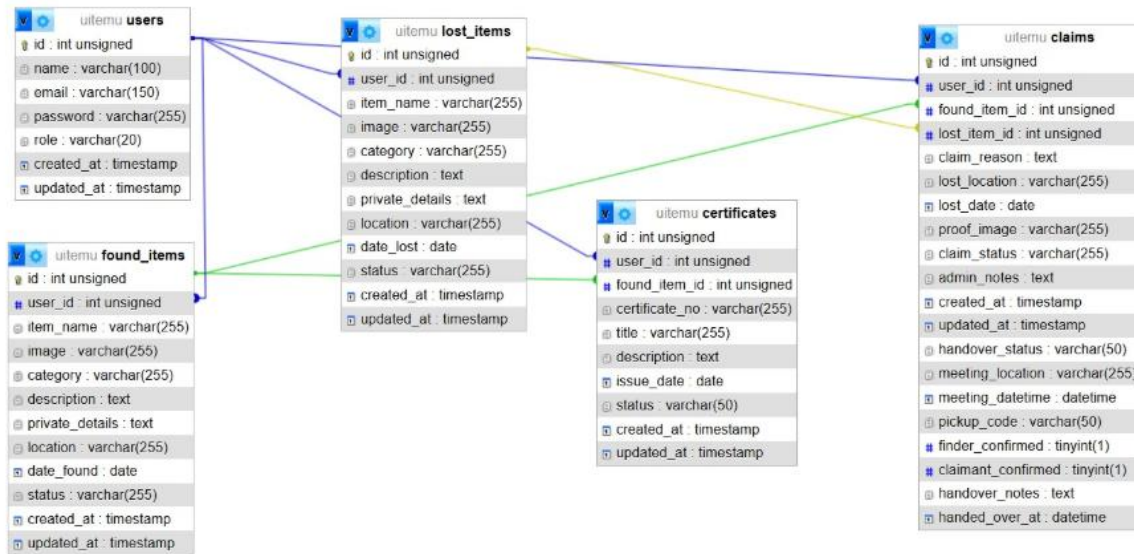


Figure 1: Entity Relationship Diagram (ERD) of UiTEMU

4.0 SYSTEM REQUIREMENTS

4.1 Software

UiTEMU software requires installation of following software to function:

CATEGORY	SOFTWARE	DESCRIPTION
Operating System	Windows 10/11	Platform used to develop and host the system.
Web Server Stack	Laragon 2026 v8.6.1	Local server package that bundles Apache, MySQL, and PHP together.
Back-End Framework	CakePHP	Handles the application's core logic following the MVC architecture.
CSS Framework	Bootstrap	Provides ready-made styling components for a responsive interface.
Database Server	MySQL 8.4.3	Stores and manages all data used within the system.
PHP Runtime	PHP 8.3.30	Executes server-side scripts that power the application.
Package Manager	Composer 2.9.7	Manages and installs the PHP libraries needed by the project.
Web Browser	Google chrome 150.0.7871.116	Used to test and access the system, in line with course requirements.
Text Editor	Visual Studio Code 1.128.1	Main tool used to write and manage the source code.

4.2 Version Compatibility

Here are the minimum and required UiTEMU versions that our environment must support to run UiTEMU correctly.

COMPONENT	MINIMUM VERSION	RECOMMENDED VERSION
PHP	8.1	8.3.30
CakePHP	Compatible with PHP 8.1+	-
MySQL	5.7 or MariaDB 10.3+	8.4.3
Composer	2.x	2.9.7
Bootstrap	5.x	-
Google Chrome	Latest version recommended	150.0.7871.116
Visual Studio Code	Latest version recommended	1.128.1

4.3 Dependencies

The UiTEMU system utilizes these third-party libraries and packages:

4.3.1 Core Dependencies

PACKAGE	VERSION	PURPOSE
cakephp/cakephp	^4.5.0	Core framework that structures the application using the MVC pattern.
cakephp/authentication	^2.11	Manages user sign-in, sign-out, and session handling.
cakephp/migrations	3.7	Tracks and applies changes to the database structure over time.
Cakephp/plugin-installer	^1.3	Automates the setup of CakePHP plugins during installation.
dompdf/dompdf	^3.1	Converts HTML content into downloadable PDF files.
friendsofcake/cakepdf	4.1	Connects CakePHP with PDF rendering engines such as DomPDF.
mobiledetect/mobiledetectlib	3.74	Identifies the user's device type to support responsive layouts.

4.3.2 Development Dependencies

PACKAGE	VERSION	PURPOSE
cakephp/bake	^2.9	Speeds up development by auto-generating boilerplate code.
cakephp/debug_kit	^4.9	Displays debugging information during local development.
cakephp/cakephp-codesniffer	^4.5	Enforces consistent coding standards across the codebase.
phpunit/phpunit	^9.6	Runs automated test to verify application functionality.
josegonzalez/dotenv	^4.0	Reads configuration values from the .env file into the app.

4.3.3 Frontend Libraries (CDN-based)

LIBRARY	VERSION	PURPOSE
Bootstrap	5.3.0	Supplies the layout grid and UI components for a responsive design.
Font Awesome	6.5.0	Adds scalable icons throughout the interface.
Chart.js	Latest	Generates visual charts for displaying system data.

5.0 USER INTERFACE OVERVIEW (LAYOUT, NAVIGATION)

UiTEMU features a clean, card-based interface that adapts its content based on the logged-in user's role, while keeping the same overall navigation structure for both students and administrators. This ensures that once a user is familiar with the layout, moving between modules feels consistent no matter which role they are using.

5.1 Overall Layout Structure

The system uses a consistent layout throughout all pages with the following components:

1. **Sidebar** – a fixed navigation panel on the left containing the UiTEMU logo and links to Dashboard, Items, Claims, Certificates, and Users (Users only appear for admin accounts).
2. **Header Bar** – sits at the top of the main content area and displays the logged-in user's name, role badge (ADMIN/USER), and a Logout link.
3. **Main Content Area** – displays the current page's content (statistics cards, tables, or forms) with consistent spacing and organization.

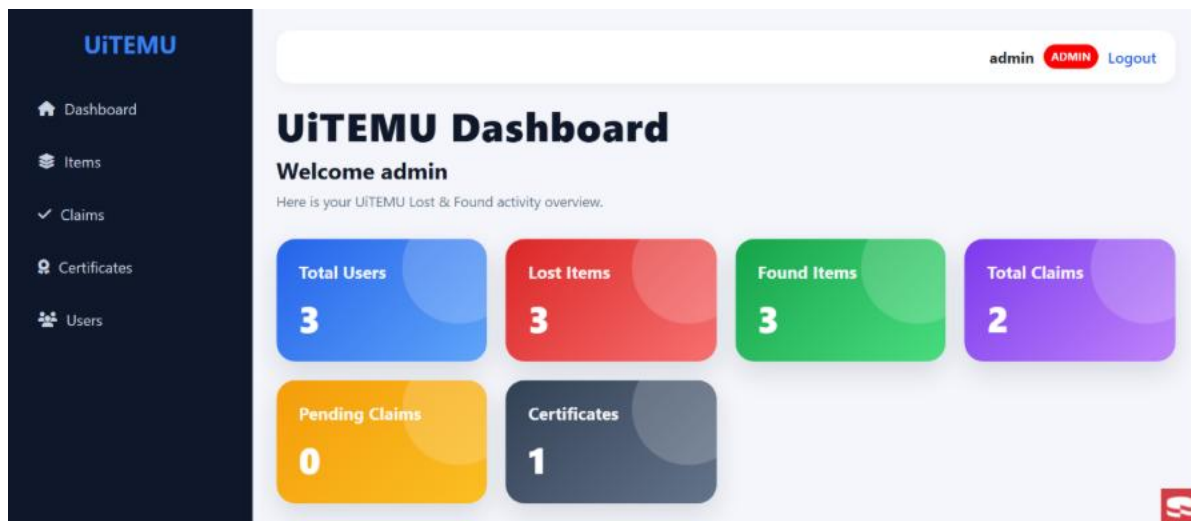


Figure 2: Admin Dashboard



Figure 3: User Dashboard

5.2 Color Scheme and Branding

The system follows a consistent color scheme, sampled directly from the interface:

ELEMENT	COLOR	HEX CODE
Sidebar / Primary Dark	Dark Navy	#0F172A
Background	Light Grey	#F5F6FB
Lost Items / Danger	Red	#E94949
Found Items / Success	Green	#32C368
Claims	Purple	#A365F6
Pending / Warning	Orange	#F8AA14
Certificates	Slate Grey	#46556C
Users	Blue	#3F81F1

These colors are applied consistently across dashboard summary cards, status tags, and buttons, so users can quickly associate a color with its record type (e.g. red for lost items, green for found items) regardless of which page they are viewing.

5.3 Navigation System

Navigation is handled entirely through the sidebar, which adapts slightly based on the user's role:

STUDENT NAVIGATION MENU	ADMINISTRATOR NAVIGATION MENU
Dashboard	Dashboard
Items	Items
Claims	Claims
Certificates	Certificates
	Users

Module-specific actions (e.g. “+ Report Lost Item,” “+ New Claim,” “+ New Certificate,” “+ New User”) are placed as buttons at the top of each page rather than nested inside the sidebar, keeping navigation simple while placing relevant actions closer to where they are used.

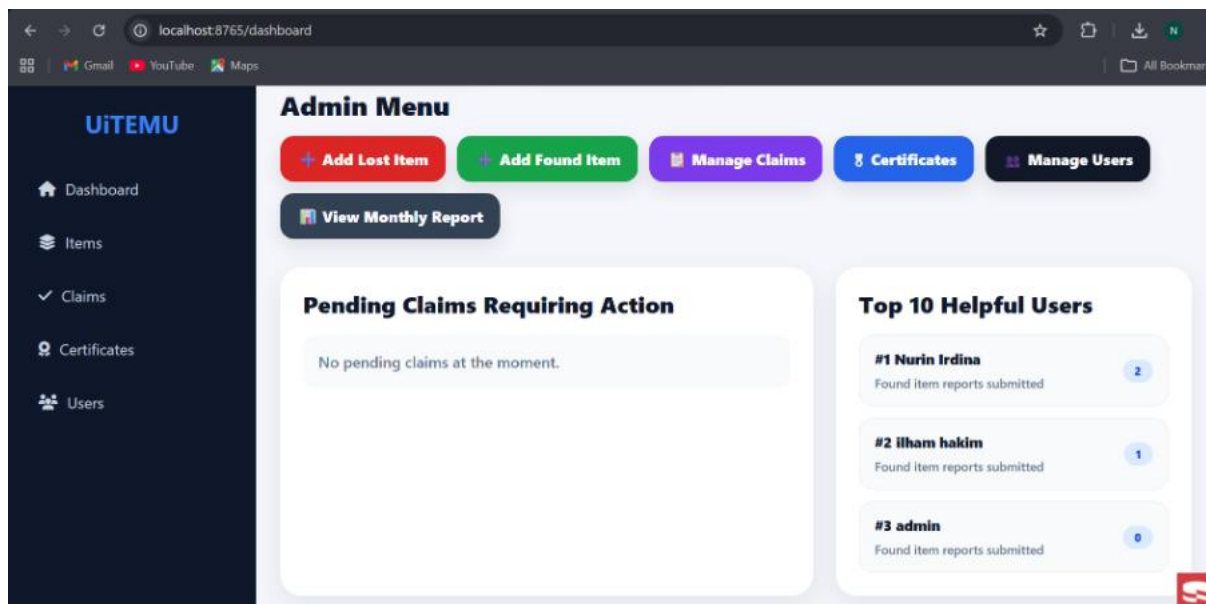


Figure 4: Admin Menu Shortcuts Shown on the Admin Dashboard

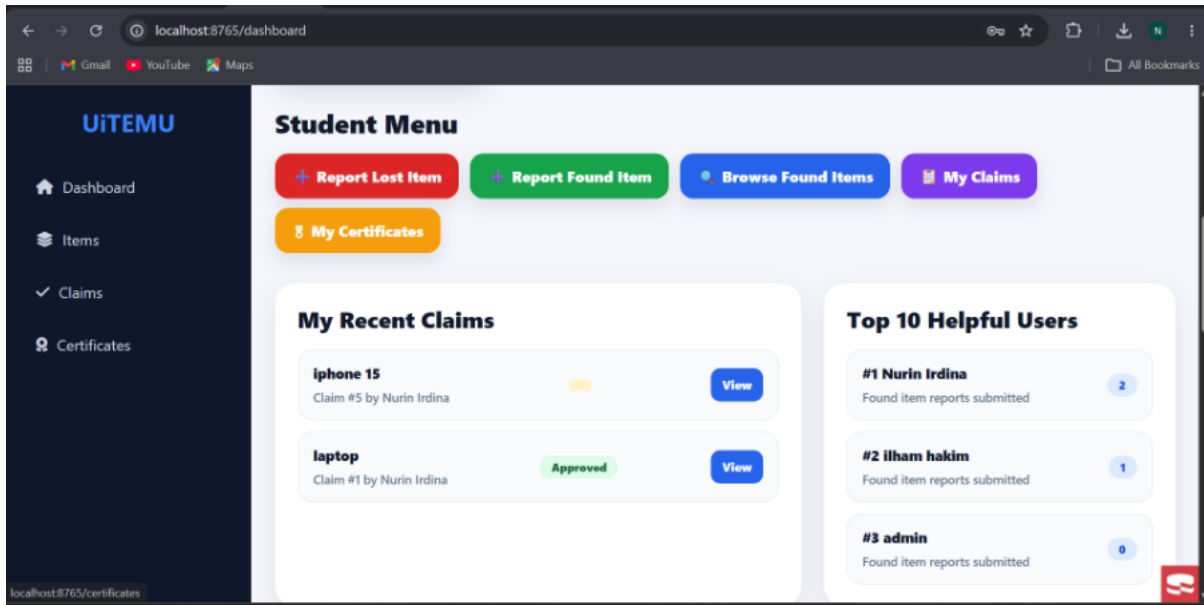


Figure 5: Student Menu Shortcuts Shown on the Student Dashboard

5.4 Page Layout Components

Most pages in UiTEMU are built from the same recurring components:

1. Header bar – user's name, role badge, and Logout link
2. Page title and description – a bold heading followed by a short explanation of the page's purpose
3. Action buttons – primary buttons placed top-right (e.g. “+ New Claim”)
4. Card-based content – items, claims, and dashboard statistics shown as individual cards
5. Tables with pagination – used in Certificates and Users Management
6. Charts – bar, pie, and donut charts summarising activity data
7. Two-column forms – an instructional panel on the left, input fields on the right

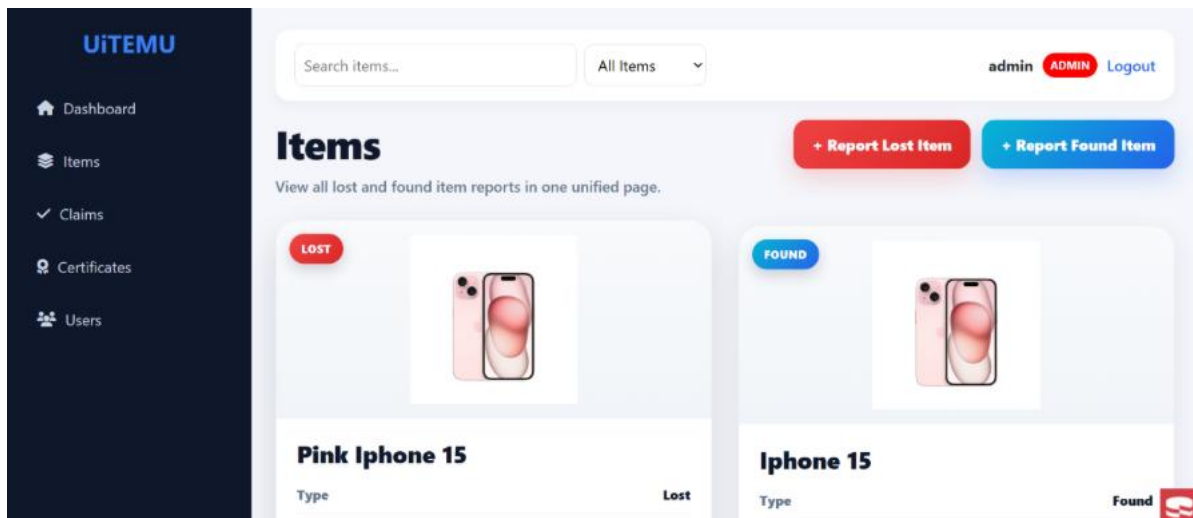


Figure 6: Items Module Showing the Search Bar, Filter Dropdown, and Card Layout

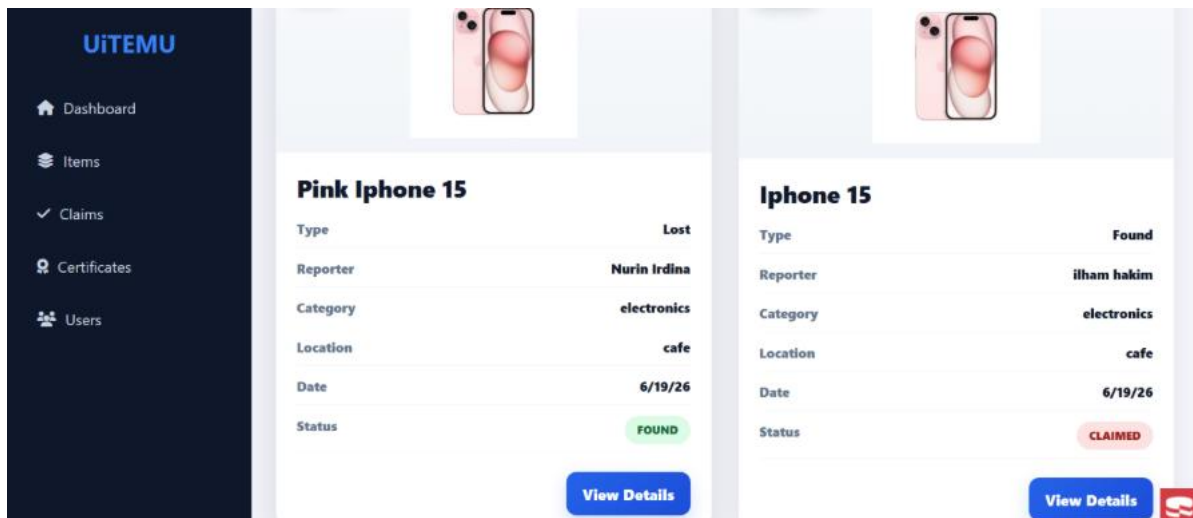


Figure 7: Report Lost Item Form Showing the Two-column Form Layout with An Instructional Panel

5.5 Mobile Responsive Design

UiTEMU's layout is built from standard responsive components to a fixed sidebar, fluid content width, and card/table elements that adapt to a single-column view on smaller screens, as shown below on a mobile width viewport. Dedicated mobile specific testing across a wider range of screen sizes and devices was not part of the current project scope and is noted as a recommendation for future enhancement.

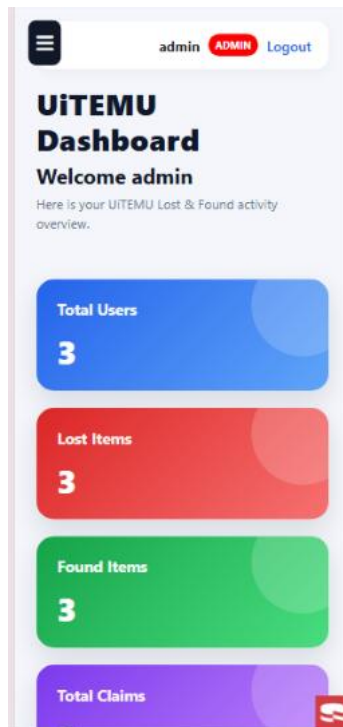


Figure 8: Admin Dashboard Rendered at Mobile Viewport Width, Showing the Layout Collapsing to a Single Column

6.0 FEATURES AND FUNCTIONALITIES

This section describes each module of UiTEMU in detail, covering both the student (user) view and the administrator view where the functionality differs.

6.1 Authentication Module

Access to UiTEMU begins with a split-panel authentication screen shared by the Sign in and Create Account pages, keeping the branding and messaging consistent while the form fields on the right change to match the task.

1. Sign In – Email Address and Password fields, a Login button, and a link to the registration page for new users
2. Create Account – Full Name, Email Address, and Password fields, a Sign-Up button, and a link back to the sign-in page for existing users
3. Left-hand panel – UiTEMU logo and a short welcome message that changes depending on whether the user is signing in or creating an account

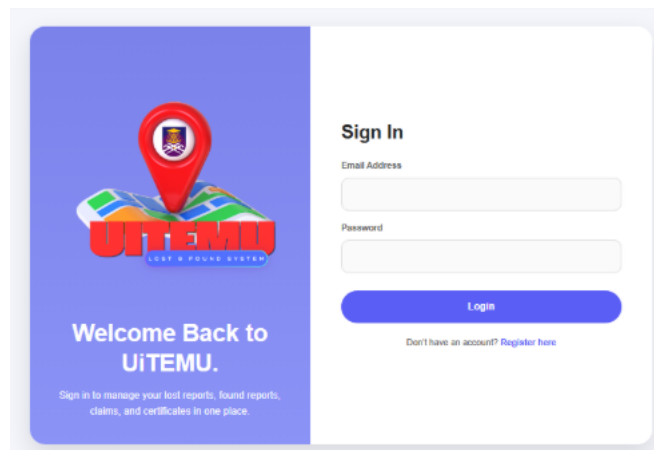


Figure 9: Sign In Page with The Branded Left Panel and Email/Password Fields

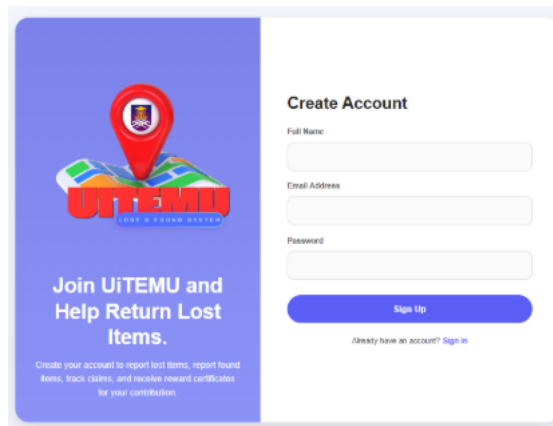


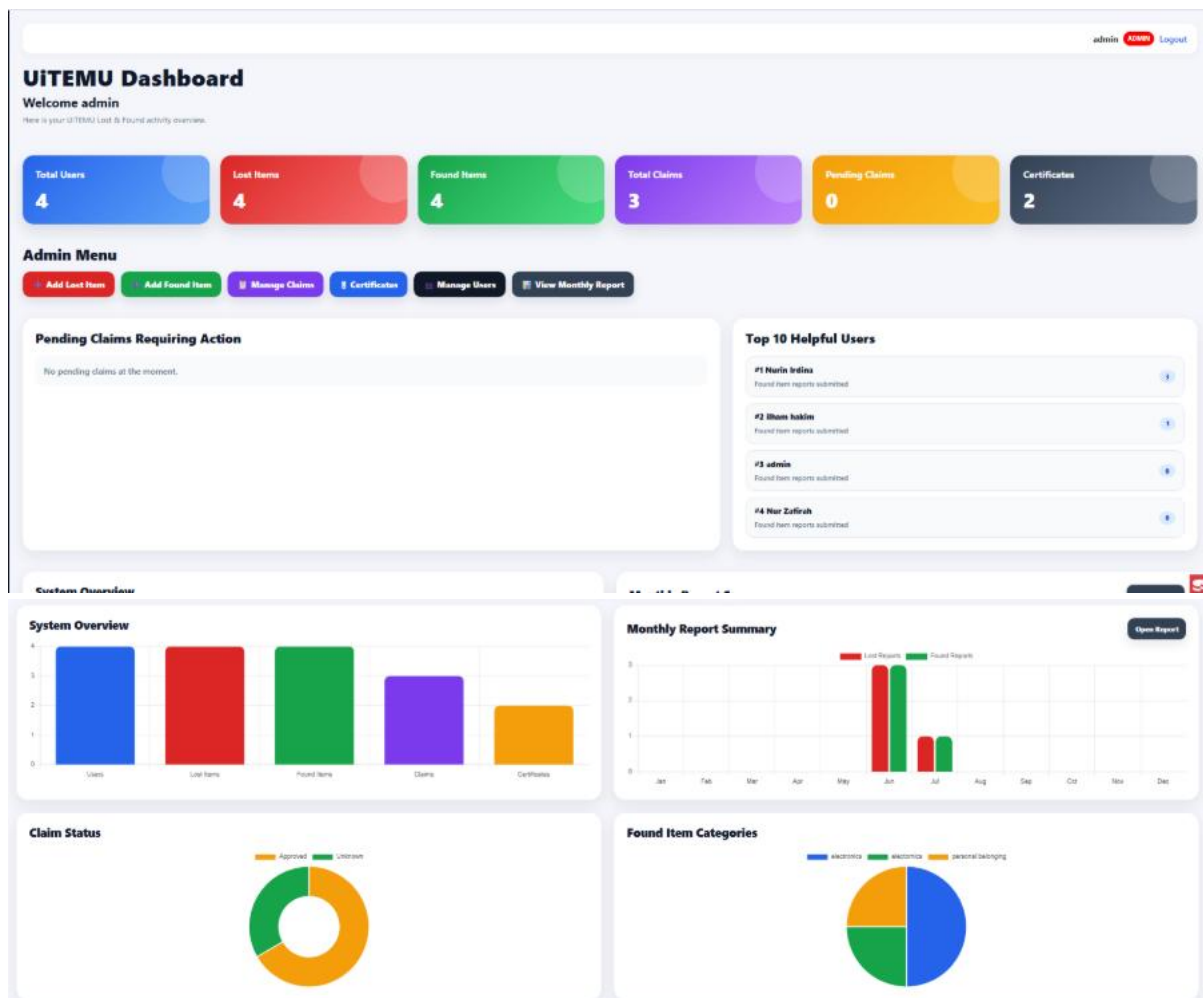
Figure 10: Create Account Page Shown After Switching from Sign In

6.2 Dashboard Module

Admin Dashboard

The Admin Dashboard gives administrators a system-wide overview of the platform.

1. Summary cards: Total Users, Lost Items, Found Items, Total Claims, Pending Claims, Certificates.
2. Admin Menu: Add Lost Item, Add Found Item, Manage Claims, Certificates, Manage Users, View Monthly Report.
3. Pending Claims Requiring Action panel.
4. Top 10 Helpful Users leaderboard.
5. System Overview bar chart and Monthly Report Summary chart.



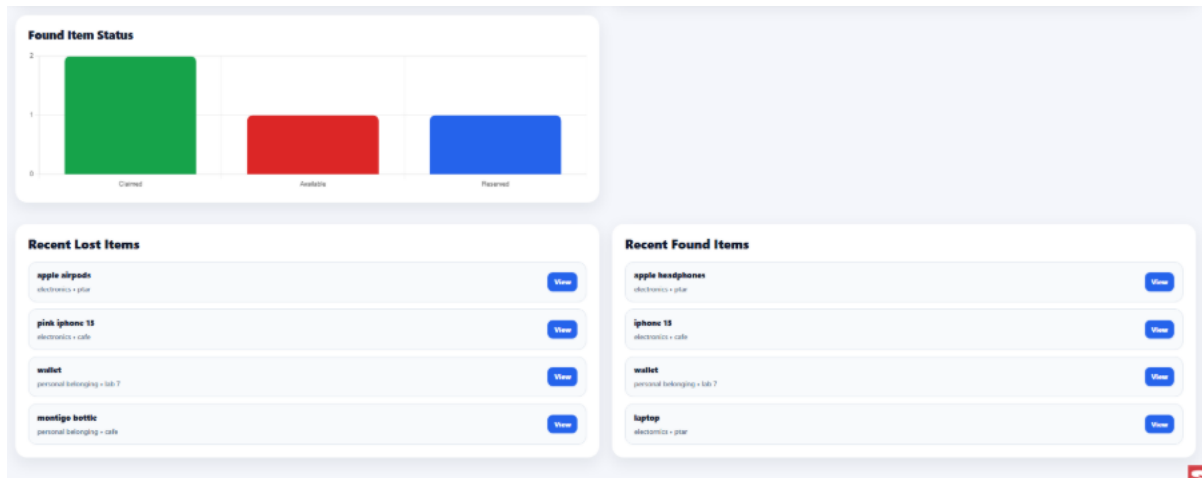


Figure 11: Admin Dashboard's Overview

Student Dashboard

The Student Dashboard mirrors this layout but scopes every statistic to the logged-in student.

1. Summary cards: My Lost Reports, My Found Reports, My Claims, Pending Claims, My Certificates.
2. Student Menu: Report Lost Item, Report Found Item, Browse Found Items, My Claims, My Certificates.
3. My Recent Claims and My Recent Lost/Found Reports lists.
4. My Activity Overview chart, Claim Status donut chart, Found Item Category and Status charts.

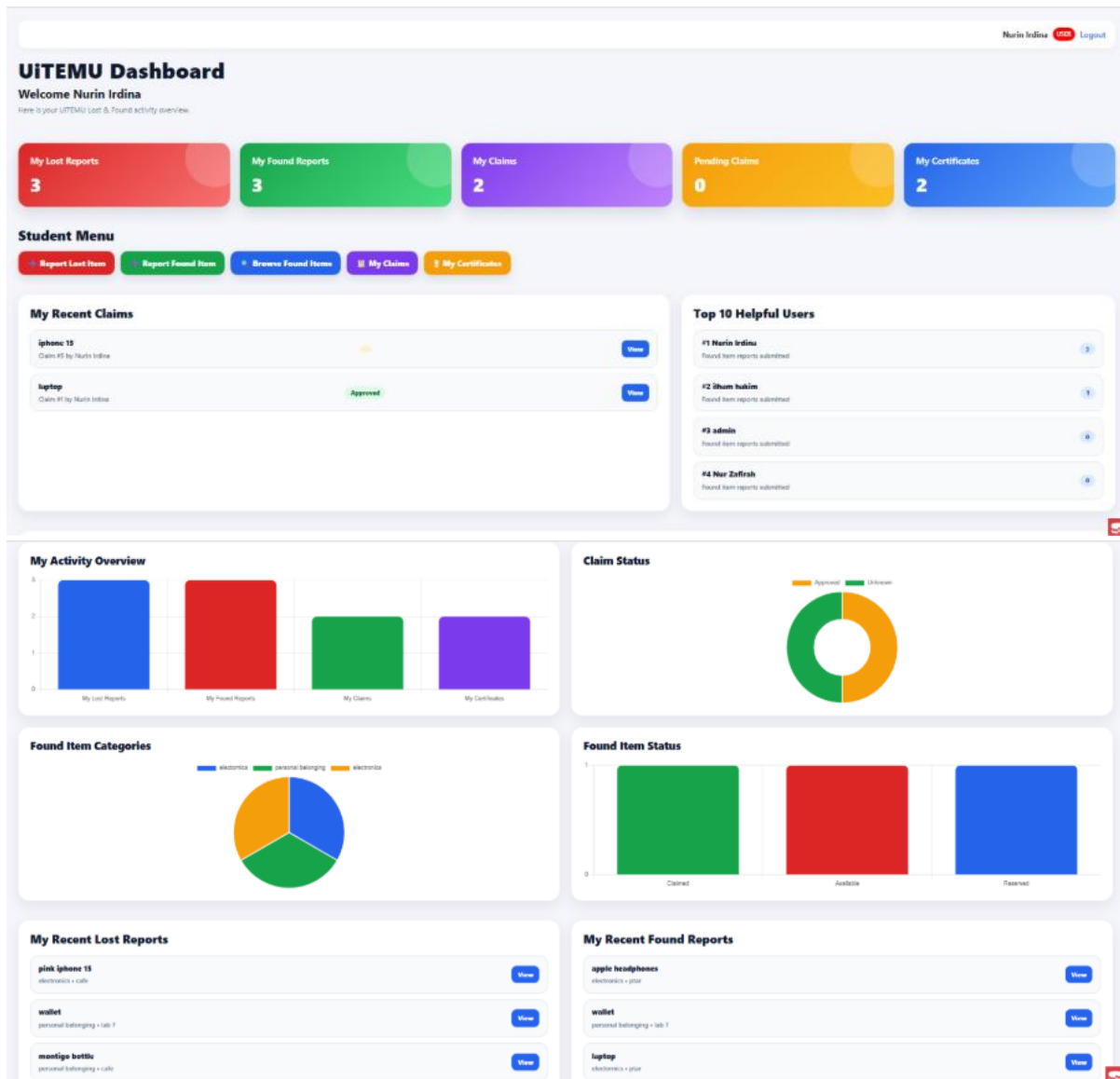


Figure 12: User Dashboard's Overview

6.3 Items Module

The Items module presents lost and found reports together on one unified page.

1. Search bar and “All Items” filter dropdown.
2. Item cards showing image, name, and a LOST or FOUND tag.
3. “View Details” button revealing Type, Reporter, Category, Location, Date, and Status.
4. Report Lost Item and Report Found Item forms, both using a two-column layout with tips on the left and form fields (Item Name, Category, Location, Date, Description, Private Verification Details, Upload Image) on the right.

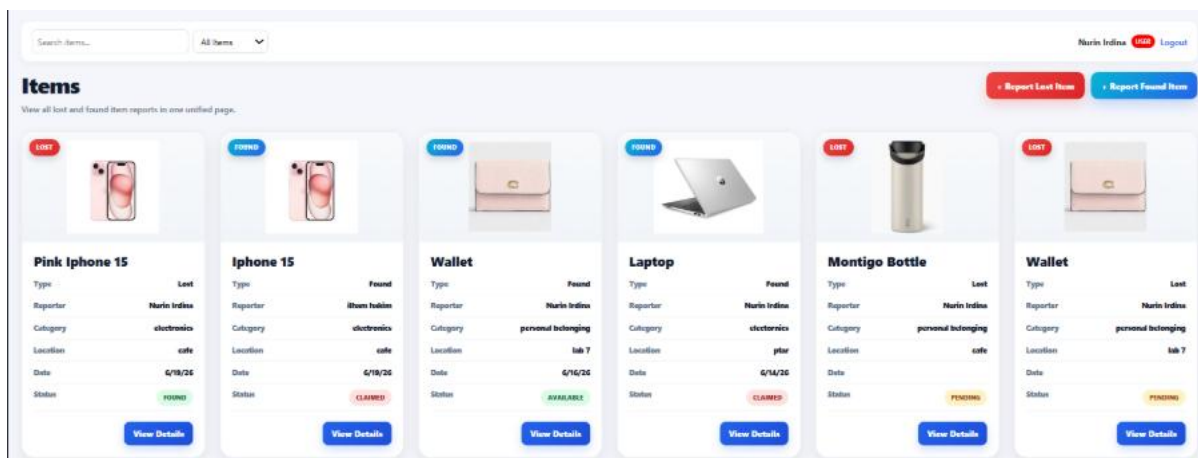


Figure 13: Item Details View Showing Type, Reporter, Category, Location, Date, and Status

Report Lost Item

Fill in the details carefully so the admin can match your lost item with any reported found item.



Lost Something?

Submit your lost item report with clear details. This helps UITEMU verify ownership when someone finds your item.

- Use a clear item name.
- Add the location where you lost it.
- Include private details only the owner knows.
- Upload an image if you have one.

Lost Item Details

Private verification details will help admin confirm that the item really belongs to you.

Item Name Example: iPhone 15, wallet, bottle	Category Example: electronics, personal belonging
Lost Location Example: cafe, library, lab 7	Date Lost dd/mm/yyyy

Description

Describe the item generally. Example: pink phone, black wallet, silver laptop.

Private Verification Details

Example: lock screen picture, sticker, scratch, hidden mark, special feature.

Upload Image

No file chosen

Figure 14: Report Lost Item Form

Report Found Item

Submit details of the item you found so the owner can identify and claim it safely.



Found an Item?

Report the item clearly. This helps the owner search for it and helps admin verify ownership before returning it.

- Use a clear item name.
- Add the exact found location.
- Include private details for verification.
- Upload a clear image of the item.

Found Item Details

Private verification details should not be obvious to everyone. They help admin confirm the real owner.

Item Name Example: iPhone 15, wallet, bottle	Category Example: electronics, personal belonging
Found Location Example: cafe, library, lab 7	Date Found dd/mm/yyyy

Description

Describe the item generally. Example: pink phone, black wallet, silver laptop.

Private Verification Details

Example: brand, defect, label, mark, item contents, hidden details.

Upload Image

No file chosen

Figure 15: Report Found Item Form

6.4 Certificates Module

This module manages appreciation certificates issued to users who help return found items. The table lists No., User, Certificate No. (auto-generated, e.g. CERT-20260619-001), Reported Item, Issue Date, Status, Created, Updated, and Actions.

1. Admin: View, Download PDF, Edit, Delete, plus a “+ New Certificate” button.
2. Student: View and Download PDF for their own certificates only.

The screenshot displays the UITEMU Certificates module interface. On the left is a dark sidebar with navigation links: Dashboard, Items, Claims, Certificates, and Users. The main content area has a header with 'admin ADMIN Logout' and a '+ New Certificate' button. Below the header is the 'Certificates' section with a subtitle 'Manage and view generated appreciation certificates.' and a table of certificates.

No	User	Certificate No	Reported Item	Issue Date	Status	Created	Updated	Actions
1	Nurin Irdina	CERT-20260619-001	laptop	6/13/26	Issued	6/13/26, 10:17 AM	6/19/26, 11:47 AM	View Download PDF Edit Delete

Page 1 of 1 showing 1 record(s) out of 1 total

Below the table is the 'Create Certificate' form. It includes a 'Certificate Details' section with instructions and a 'New Certificate' section with fields for Recipient User, Found Item, Certificate Number, Issue Date, Certificate Title, Description, and Status. The 'Create Certificate' button is at the bottom.

Figure 16: Certificates Table (Admin View)

UITEMU

Dashboard

Items

Claims

Certificates

Nurin Irdina

USER

Logout

Certificates

Manage and view generated appreciation certificates.

No	User	Certificate No	Reported Item	Issue Date	Status	Created	Updated	Actions
1	Nurin Irdina	CERT-20260619-001	laptop	6/13/26	Issued	6/13/26, 10:17 AM	6/19/26, 11:47 AM	View Download PDF

Page 1 of 1, showing 1 record(s) out of 1 total

Certificate Details

View certificate information and download the PDF copy.

Certificate No: CERT-20260619-001

CERTIFICATE OF APPRECIATION

This certificate is proudly presented to

Nurin Irdina

In recognition of honesty, responsibility, and contribution to the UITEMU Lost & Found System.

Reported Found Item:
laptop

Issued Date: 6/13/26

Authorized Signature
UITEMU Lost & Found System

Certificate Information

Recipient:
Nurin Irdina

Reported Item:
laptop

Status:
Issued

Created At:
6/13/26, 10:17 AM

Certificate No:
CERT-20260619-001

Issue Date:
6/13/26

Certificate ID:
#1

Updated At:
6/19/26, 11:47 AM

Download PDF

Issued

Actions

Back to Certificates

Download PDF

Figure 17: Certificates Table (Student View), Limited to View and Download PDF

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6.5 Claims and Handover Management Module

This module coordinates the process of claiming a found item and confirming its return.

1. Found item cards showing Found Item ID, Found By, Category, Location, and Item Status.
2. Claim Details panel showing Claim ID, Claimed By, Lost Item, Claim Status, and Handover status (e.g. Not Arranged, Approved, Disputed).
3. Finder actions (student view): Confirm Handed Over, Confirm Received, Report Problem.
4. Admin oversight: View, Edit, and Delete on any claim, used mainly to resolve disputes.

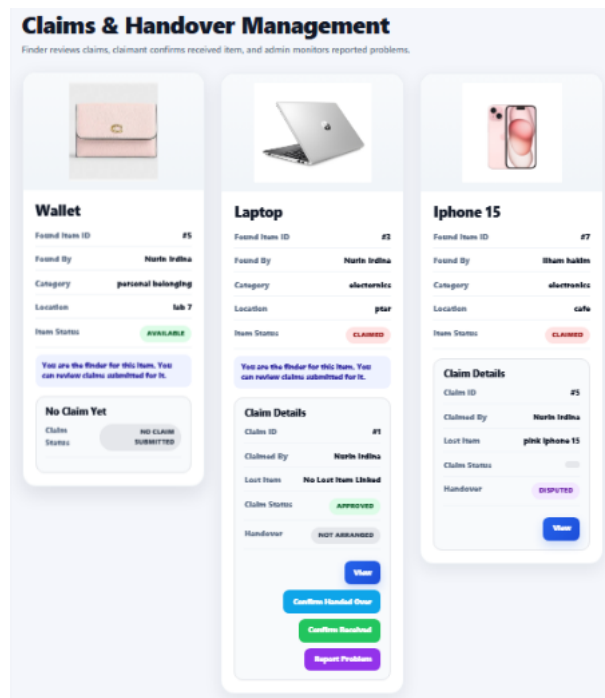


Figure 18: Claims & Handover Management (Student/Finder)

Claims & Handover Management

Finder reviews claims, claimant confirms received item, and admin monitors reported problems.



iPhone 15

Found Item ID #7

Found By **ilham hakim**

Category **electronics**

Location **cafe**

Item Status **CLAIMED**

Claim Details

Claim ID #5

Claimed By **Nurin Irdina**

Lost Item **pink iphone 15**

Claim Status ☐

Handover **DISPUTED**

[View](#) [Edit](#) [Delete](#)



Wallet

Found Item ID #5

Found By **Nurin Irdina**

Category **personal belonging**

Location **lab 7**

Item Status **AVAILABLE**

No Claim Yet

Claim Status **NO CLAIM SUBMITTED**



Laptop

Found Item ID #3

Found By **Nurin Irdina**

Category **electronics**

Location **ptar**

Item Status **CLAIMED**

Claim Details

Claim ID #1

Claimed By **Nurin Irdina**

Lost Item **No Lost Item Linked**

Claim Status **APPROVED**

Handover **NOT ARRANGED**

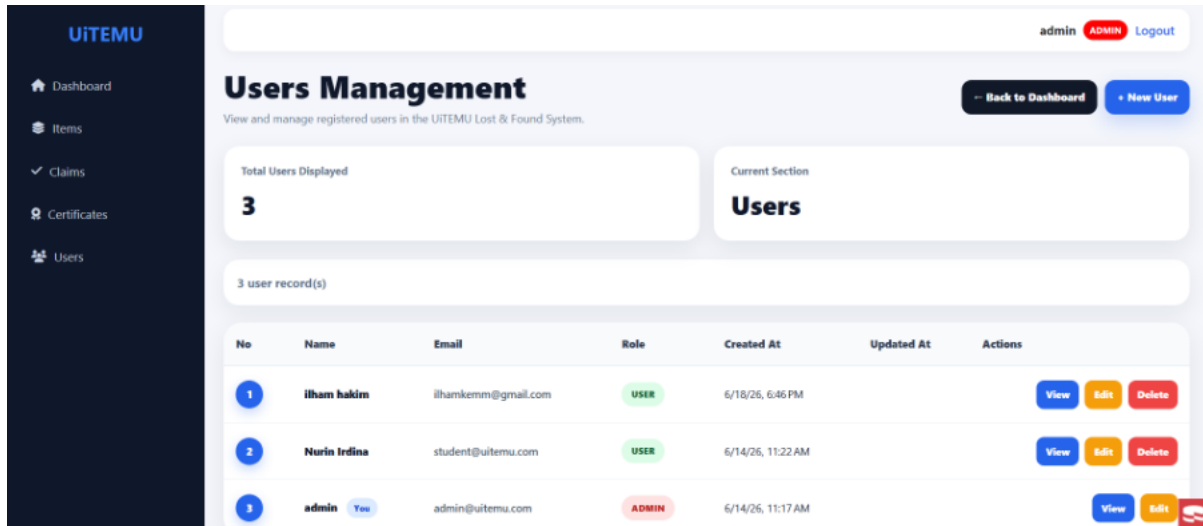
[View](#) [Edit](#) [Delete](#)

Figure 19: Claim Details Panel with View/Edit/Delete Actions (Admin View)

6.6 User Management Module (Admin Only)

Accessible only to admin accounts, this module lists all registered users.

1. Table columns: Name, Email, Role, Created At, Updated At, Actions.
2. Role tags (ADMIN/USER) and a total user count.
3. “+ New User” button, plus View/Edit/Delete on existing accounts.



No	Name	Email	Role	Created At	Updated At	Actions
1	Iham hakim	ihamkemm@gmail.com	USER	6/18/26, 6:46 PM		View Edit Delete
2	Nurin Irdina	student@uitemu.com	USER	6/14/26, 11:22 AM		View Edit Delete
3	admin	admin@uitemu.com	ADMIN	6/14/26, 11:17 AM		View Edit Delete

Figure 20: Users Management Table Listing All Registered Accounts

6.7 Reports and Analytics Module (Admin Only)

Also restricted to admin, this module aggregates system-wide data for monitoring trends.

1. System Overview bar chart comparing Users, Lost Items, Found Items, Claims, and Certificates.
2. Monthly Report Summary comparing lost vs. found reports across the year, expandable via “Open Report” / “View Monthly Report”.
3. Claim Status and Found Item Category breakdowns.

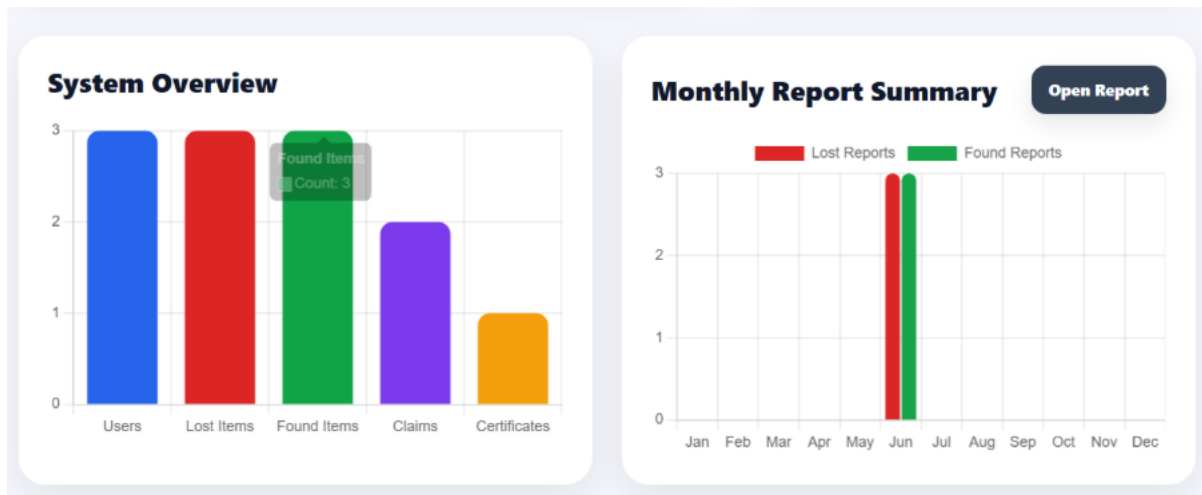


Figure 21: System Overview and Monthly Report Summary Charts on the Admin Dashboard

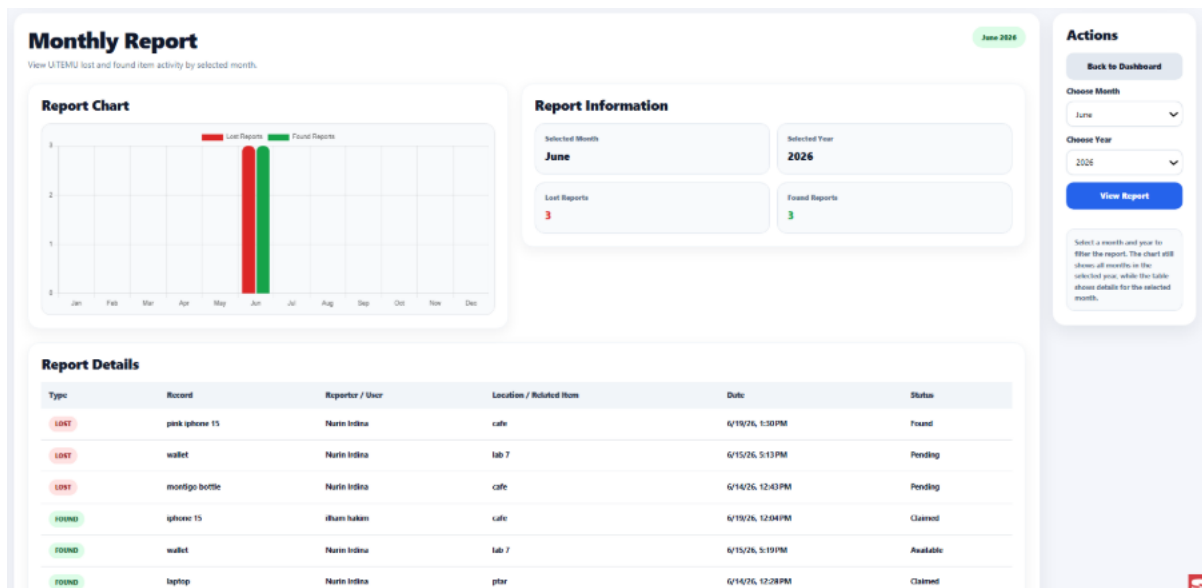


Figure 22: Monthly Report View

7.0 WORKFLOW OF FORM

7.1 CRUD Workflow Diagram

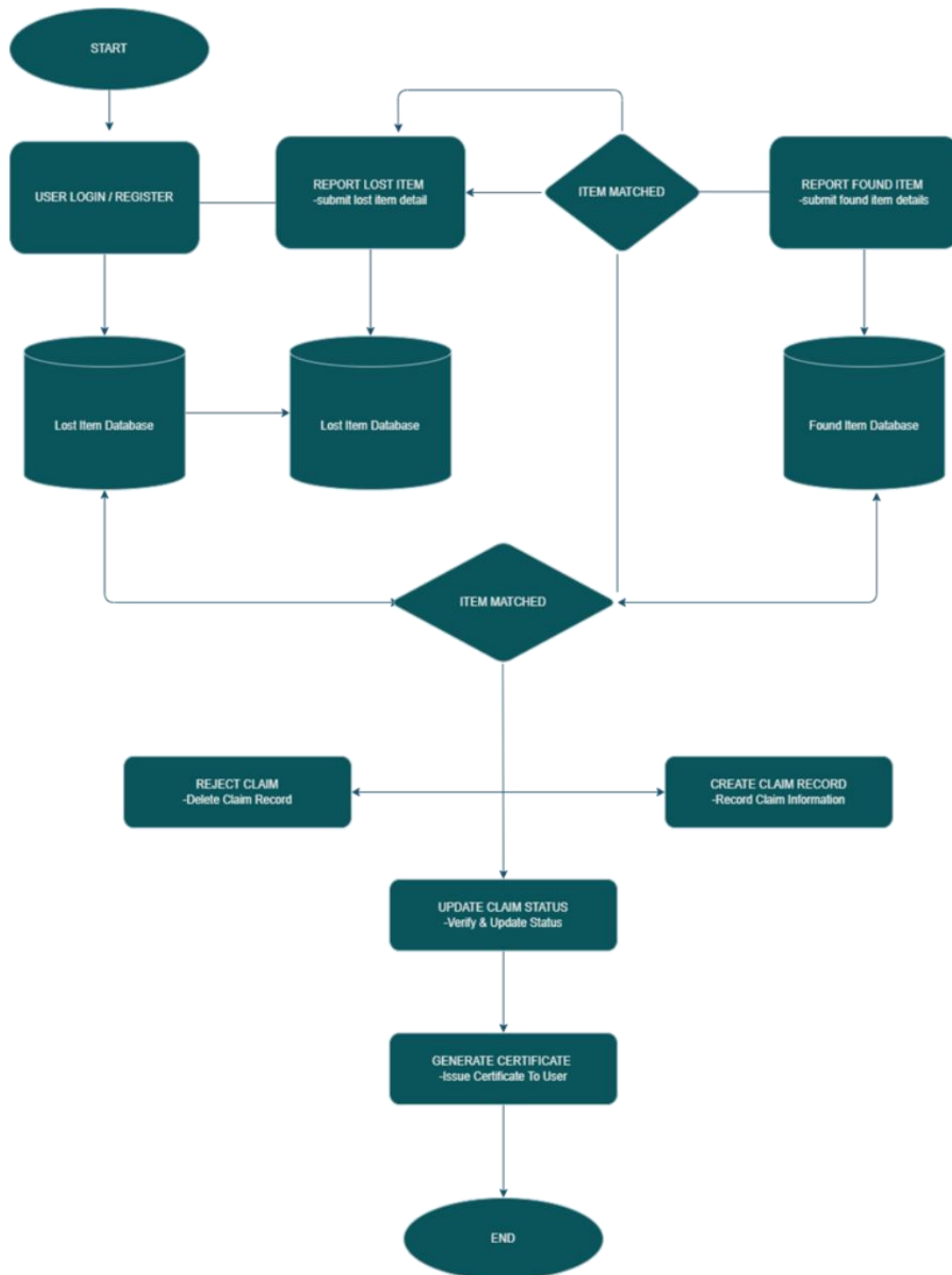


Figure 23: CRUD Workflow Diagram Lost & Found System

7.2 CRUD Process Description

CREATE – Report for lost or found items.

Step: STUDENT: Fill in and enter the details of the lost or found item.

- **Action:** The user goes to the item navigation section and selects Report Lost Item or Report Found Item and completes all the required sections (item name, category, lost location, date lost, description, private verification detail, upload image).
- **System:** Data will be sent to the server and stored in the Found Item or Lost Item table.
- **Default:** Item status becomes pending.
- **Result:** New records will appear in the database.

READ – Display list of claims and items

Step: ADMIN: Check the list of found or lost items.

- **Action:** Admin clicks on Items or Claim.
- **System:** Runs the command to retrieve data from the table.
- **Logic:** The display will be a reference before further action is continued (making a claim and checking the status).
- **Result:** Data will be shown in the list along with the status.

UPDATE – Review and submission process

Step: ADMIN: Check the item claim.

- **Trigger:** Admin will select the item claim from the list that is still pending.
- **Decision:** Accept or reject the item.
- **Path A (Accept):** The system will update the claim status to be approved, add Handover status and meeting location.
- **Path B (Reject):** The system will update the claim status to be rejected.

Step: STUDENT: Need to confirm receipt of the item at claimant confirmed.

- **Result:** The item status will be updated, and a certificate will be generated if successful.

DELETE – Withdrawing an invalid item claim or record

Step: STUDENT: Cancel the item claim.

- **Decision:** Status that are still pending.
- **Yes:** The system will delete the item claim record from the claims list.
- **No:** If the status is approved or rejected, no cancellation will be possible.

Step: ADMIN: Delete the item record that has been found or lost after the claim is made.

- **Result:** The record will be deleted or marked as inactive.

GENERATE – Submitting a certificate

Step: SYSTEM: Generate a certificate after the claim is made.

- **Trigger:** It will be automatic after the claim status is approved.
- **Output:** New record in the list of certificates with the certificate number.
- **Result:** Certificate can be downloaded as PDF.

7.3 CRUD Operation for Other Modules

USER CRUD

OPERATION	ACTION	RESULT
CREATE	Student registers an account	User record added to list
READ	Admin can view profile	User information will be displayed
UPDATE	User updates profile and admin change's role	User data has been updated
DELETE	Admin deletes inactive account	User record will be deleted

LOST ITEM CRUD

OPERATION	ACTION	RESULT
CREATE	Student reports lost items	Item record appears in lost item list
READ	Admin and user view list of lost items	Lost item list is shown
UPDATE	Admin will change status of found item	Item status changes
DELETE	Admin will delete record of claimed item	Item record will be deleted

FOUND ITEM CRUD

OPERATION	ACTION	RESULT
CREATE	User reports lost item	New item record in found items list
READ	Admin and user view list of found items	Item list will be shown
UPDATE	Admin changes status when item will be claimed	Item status will have changed
DELETE	Admin deletes claimed item record	Item record will be deleted

CLAIMS CRUD

OPERATION	ACTION	RESULT
CREATE	User claims item	Item record is out in claims list
READ	Admin reviews claim	Claim status will be shown
UPDATE	Admin changes status to approved or rejected	Status has changed
DELETE	Claim will be rejected	Record deleted

CERTIFICATE CRUD

OPERATION	ACTION	RESULT
CREATE	System generates certificate after claim is made	Certificate records will be added
READ	User views certificate	Certificate will be shown
UPDATE	Admin changes certificate status	Certificate status will change
DELETE	Admin revokes invalid certificate	Certificate record will be deleted

7.4 Team Roles and Contribution

No.	NAME	ROLE	CONTRIBUTION
1.	ADANI SOFIA BINTI MOHD ASRI	Project Leader	Distribute tasks to all members, ensure tasks are in accordance with the format, and provide ideas for creating a system.
2.	FARISHA NADZIRAH BINTI AHMAD AFANDY	System Developer	Perform tasks assigned in the report, prepare workflow diagrams, and create documentation for the CRUD and workflow sections in the report.
3.	NUR ZAFIRAH BINTI AMINULDIN	Database Developer	Design and develop the Lost & Found database structure, create and maintain all database tables (Users, Lost Objects, Found Objects, Claims, Certificates), set up foreign key associations and enforce data integrity constraints, and develop all queries that will be utilized by the application.
4.	NURIN IRDINA BINTI ABDULLAH	Programmer Developer	Design the back-end code and server-side program of the UiTEMU system, perform CRUD functionalities on all modules, integrate the system through the front-end user interface and the database, design the certificate generation functionality and test and debug the entire system.

8.0 CONTACT INFORMATION

8.1 Organization Contact

General Enquiries, Feedback & Support Users may get in touch with the System Administrator via the following methods to reach out to the system regarding any general questions, feedback, comments, or to seek assistance.

CONTACT TYPE	DETAILS
Organisation	UiTEMU Lost & Found System (UiTM Puncak Perdana)
Address	Kolej Jasmine, Laman 4, Bangunan Pentadbiran, UiTM Kampus Puncak Perdana, 40150 Shah Alam, Selangor
Phone	+603-5543 5000
Email	lostandfound@uitm.edu.my
System URL	localhost:8765 (internal campus network)
Operating Hours	Monday – Friday, 8:00 AM – 5:00 PM

8.2 Technical Support

For any technical problems with UiTEMU Lost & Found System, please contact respective tech support staff depending on the nature of the technical problem.

SUPPORT TYPE	CONTACT
System Administrator / Project Leader	Adani Sofia binti Mohd Asri 2025236312@student.uitm.edu.my
System Developer (Frontend & UI Issues)	Farisha Nadzirah binti Ahmad Afandy 2025231574@student.uitm.edu.my
Database Issues & Data Queries	Nur Zafirah binti Aminuldin 2025219844@student.uitm.edu.my
Bug Reports & Backend Issues	Nurin Irdina binti Abdullah 2025234622@student.uitm.edu.my

8.3 Academic Contact

Project has been made as a part of course IMS566 for academic inquiries about the system and for course assignment queries please contact course lecture:

CONTACT TYPE	DETAILS
Course	IMS 566 — Web-Based Information Systems
Lecturer Name	Dr. Muhammad Asyraf bin Wahi Anuar
Lecturer Email	muham6853@uitm.edu.my
Faculty	Faculty of Information Management, UiTM Puncak Perdana

9.0 CONCLUSION AND REFLECTION

9.1 Project Summary

The UiTEMU Lost & Found Management System is successfully created in order to solve an existing real issue that exists among the students at UiTM Puncak Perdana where it has been difficult to report about their loss of personal property or find those items that might have been reported as lost. Before this management system was introduced, there was no official electronic means through which students could report about losing their items or looking for any items that have been lost and have been found on the campus. It has been difficult and disorganized.

The system allows a central digital platform to facilitate three main operations:

1. Reporting Lost & Found Items

Students will find it very easy to report the lost or found items using an online form that is designed in such a way as to include all necessary data about the found items. All this data includes name of the item; category; place where the item was found; date when the item was found; description of the item; personal identification details; image upload.

2. Managing Claims and Handovers

Students will be able to make a claim for any item they found, and which belongs to them. The administrators will check and verify the claim and then approve or deny the claim depending on the provided verification data. The handover status of the item will be recorded along with the place where the handover took place.

3. The Certificate of Appreciation

To reward those students who contribute positively to our community, the system will provide a digital certificate of appreciation to any student who reports a found item which will lead to a successful claim. The certificate will be downloadable in PDF format.

Moreover, there is a real-time Dashboard showing both administrators and students their activities in one view. Administrators will see the aggregated statistics about the total number of users, lost items, found items, total claims, pending claims, and number of certificates created. For students, there will be a summary of personal activity with lost reports, found reports, claims, and personal certificates. There is also a list of the Top 10 Helpful Users as well as a bar chart showing the activity of the system over time.

The implementation of role-based access control makes sure that only necessary features are available for students and administrators. For instance, admins have exclusive access to User Management, reporting monthly activities, and managing items and claims for the whole system. For the Student role, it is possible to report items, explore all posted information, submit claims, track their own claims, and check personal certificates. Thus, the system is governed cleanly.

9.2 Technical Achievements

Through the entire process of this project, the following technical components have been achieved:

- Complete CRUD functionality within five modules: Users, Lost items, found items, Claims, and Certificates – where all operations have been done appropriately with error handling.
- Role-Based Access Control (RBAC) – where the functionalities between Administrator and Student roles have been differentiated in all parts of the system, using session-based authentication.
- Authentication – user registration and login with password hashing, session handling, and role checking for every secured path.
- Certificate Generation – where the system automatically creates a unique digital certificate in PDF format when a claim is approved without any manual effort by an administrator.
- Monthly Report Download – where the administrator can download a monthly report of the Lost & Found System, which will serve as a reference document for administration.

9.3 Challenges Faced and Solutions

CHALLENGE	SOLUTION
Role-based access and visibility-Show content, menus and features on the dashboard relevant to either an Admin or a Student.	Introduced a role-check based approach for the sessions to check the user role during login and to show or hide navigational links, statistics card views, and buttons based on the role of the logged-in user.
Generating certificate PDF, generate one formatted unique certificate number in pdf format which is sent automatically without human intervention by admin.	Created an automatic certificate creation functionality when claim status is changed to 'Approved', which generates a PDF from an HTML template designed in the system's brand colors. The certificate number is auto generated for each certificate and then stored in the Certificates table.
Logic of Claim Verification – Protecting against any fraudulent claims wherein the user tries to make a claim of something which is not his own, without creating too much of a hassle for actual owners to file claims.	In the Lost Item Report page, included a private verification field wherein the user enters an exclusive information about the item only known by the actual owner. The administrator checks whether this matches the reported item to approve or disapprove the claim.
Handling Image File Upload and storage. The service to deal with the secure handling and processing of image file uploads in such a way that it will not result in server over usage or security issues.	Created a function for validating files to check file type (JPEG, PNG only), size restrictions, and assign a unique filename on uploading to avoid naming collisions. Files are saved on the server within a specific directory, and the filename is stored in the database.

9.4 Future Improvements

The following additional features are proposed by the team to enhance the functionality of the UiTEMU Lost & Found Management System in future developments:

1. Email Notifications

Automatic email notifications to alert users regarding the posting of newfound items matching their reported lost item, change in their claim status either approved/rejected or certificate generation. It will make the system more responsive and save the effort of the users to check the system.

2. Item Matching Algorithm

Development of a smart item-matching algorithm where newly posted found items can be automatically matched with existing lost item reports depending on item name, category, description, and location. Matched items will then be highlighted for both the admin and user.

3. Live Chat or In-System Messaging

Provide the option for students who have lodged an inquiry regarding a particular found item to chat with the finder using an in-system messaging facility, thereby enabling them to work out the particulars of handing over the item without having to disclose personal information.

4. QR Code Integration

Create a specific QR code for every report of a found item such that scanning it redirects users to the details page of the item. Such QR codes may be printed on paper and placed on the university's notice board, where students who came across physical flyers could gain easy access to their respective digital reports.

5. Integration with UiTM Student Login Portal

Link the UiTEMU login system with UiTM's student portal (such as i-Student and others) so that students can use their student credentials at UiTM to log into the system without creating a new user account.

10.0 REFERENCES

CakePHP Codebase Documentation. (2026). Retrieving data & result sets: The Model layer

and custom finders. CakePHP Software Foundation. <https://book.cakephp.org/5.x/orm/retrieving-data-and-resultsets.html>

Intro to DOMPDF - light and simple PHP library to generate PDF documents. DEV

Community. (2024, April 5). <https://dev.to/prahladyeri/intro-to-dompdf-lightest-and-simplest-php-library-to-generate-pdf-documents-218n>

Staff, R. (2024, July 24). What are System Requirements? Requirements.com.

<https://requirements.com/Content/What-is/what-are-system-requirements>

What is crud? explained. Codecademy. (n.d.).

<https://www.codecademy.com/article/what-iscrud-explained>

What is Entity Relationship Diagram (ERD)? (n.d.).

<https://www.visual-paradigm.com/guide/data-modeling/what-is-entity-relationship-diagram/>